

Junye Zhou

Portfolio: moonlitjaderabbit.github.io | junye.zhou@mail.utoronto.ca

EDUCATION

University of Toronto

Toronto, ON

Bachelor of Applied Science in Computer Engineering

2025 – 2030

- Relevant coursework: CIV102 (structures), PHY180 (classical mechanics lab), Praxis I & II (engineering design)
- CIV102 matboard bridge ranked **#1 in cohort**, **#2 in EngSci Class of 2025**

PROJECTS

CIV102 Matboard Bridge (Team 602) | *Python, SolidWorks, OptiCutter*

Nov 2026 – Dec 2026

- Designed, analyzed, and fabricated a 1.25 m matboard bridge predicted to withstand 954 N before failure while meeting strict material constraints
- Led hand calculations with teammates; contributed to Python structural analysis (GitHub)
- Built and tested full-scale prototype; splice failure near 1000 N drove four design iterations (diaphragms, splice reinforcement)
- Team: Declan Chan, Junye Zhou, Navneet Saxena, Tianhao Fu

Praxis II — Showcase Team Project | *Requirements, Capstan Design*

Feb 2026 – Apr 2026

- Partner: Mississauga Aquatic Club (MSSAC); RFP focus on variable swim resistance for training
- Designed a capstan-based resistance system providing eight adjustable resistance levels (60–170 N) with approximately 3-second switching time
- Built requirement tables, material Pugh charts, and verification plans; presented at Praxis II showcase
- Team: Grant Liang, Sean Ding, Nathan Xia, Junye Zhou

Praxis II — Beta Release RFP | *Stakeholder research, technical writing*

Jan 2026 – Feb 2026

- Developed an engineering requirements proposal to solve STL Robotics' challenge of collecting loose parts from portable competition floors and sorting reusable hardware safely and efficiently
- Conducted stakeholder interviews and portable-space measurements; framed safety and ergonomics requirements
- Team: Grant Liang, Sean Ding, Nathan Xia, Junye Zhou

PHY180 Damped Pendulum Lab | *OSP Tracker, Python, data analysis*

Oct. 2025 – Jan. 2026

- Solo lab report: filmed pendulum at 60 fps; tracked motion frame-by-frame in OSP Tracker
- Analyzed period vs. angle, damping, length scaling, and Q vs. length; $Q \approx 109$ (two methods agreed)
- Documented uncertainty sources and compared results to course theory

Praxis I — Alpha Release Project | *Proxy testing, Pugh analysis*

Sept 2025 – Dec 2025

- Designed an Alpha Release solution addressing conflicting sleep and study lighting requirements in shared residence rooms
- Diverged on four concepts; ran proxy illuminance tests in a real dorm; converged with Pugh charts
- Recommended foldable bed dome cover targeting <1 lux at the sleeper's eyes
- Team: Jiaxu Zhao, Junye Zhou, Jocelyn Li, Audrey Lin

TECHNICAL SKILLS

Programming: Python, Java, C

Engineering Software: SolidWorks, OptiCutter, OSP Tracker

Design methods: Requirements (NGO), Pugh matrices, proxy testing, stakeholder interviews, concept selection